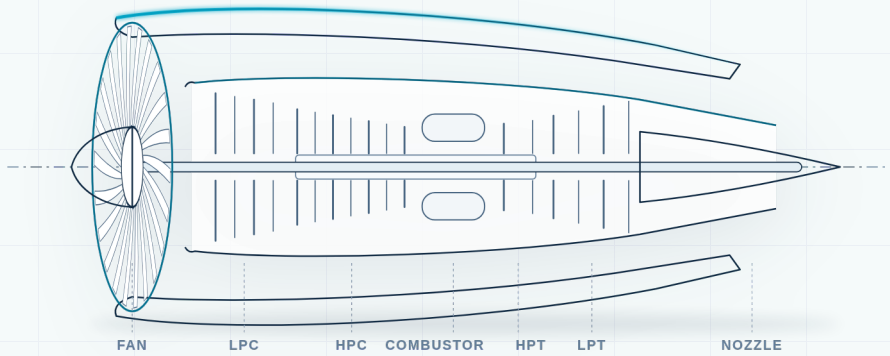
An Uncertainty-Aware, Explainable and Human-Reviewed Framework for Remaining-Useful-Life-Based Maintenance Prioritisation

Evidence from the NASA C-MAPSS FD001 Turbofan Dataset



Illustrative turbofan schematic — not a validated engine simulation

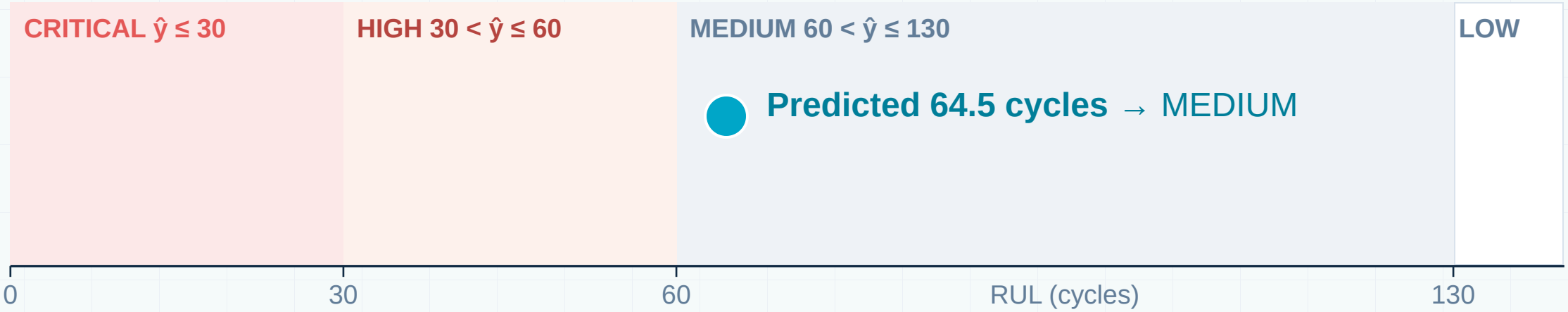
RESEARCH QUESTION

Can RUL evidence support maintenance prioritisation without hiding uncertainty or removing human authority?

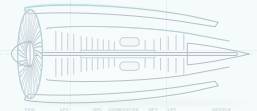
Ranitabh Mallick · Student ID GH1047661
MSc Data Science, Artificial Intelligence and Digital Business
Supervisor: Dr Hiba Khalid
GISMA University of Applied Sciences · Berlin, September 2026

A point estimate can hide maintenance risk

Class thresholds are stipulated research assumptions (Table 4.4) — not approved aircraft-maintenance rules



Source: Engine 18, official endpoint (Table 5.9); classes (Table 4.4) **Backup: triage classes ↗**



The gap is governed integration, not a new algorithm

Established components

RUL prediction
Saxena et al. (2008a); Ramasso and Saxena (2014)

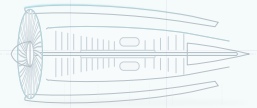
Predictive uncertainty
Angelopoulos and Bates (2023); Javanmardi and Hüllermeier (2023)

Explainability
Lundberg and Lee (2017); Vollert, Atzmueller and Theissler (2021)

Maintenance prioritisation
Jardine, Lin and Banjevic (2006); ISO 17359 (ISO, 2018a)

Human oversight
Parasuraman, Sheridan and Wickens (2000); Shneiderman (2020)

Targeted narrative review (§3.1–3.7, Table 3.1) — not a systematic review **Backup: references ↗**



Separation by engine, not by cycle row

Secondary, simulated NASA C-MAPSS FD001 · high-pressure-compressor degradation · one operating condition

Grouped for display; roles came from a seed-42 engine-level shuffle

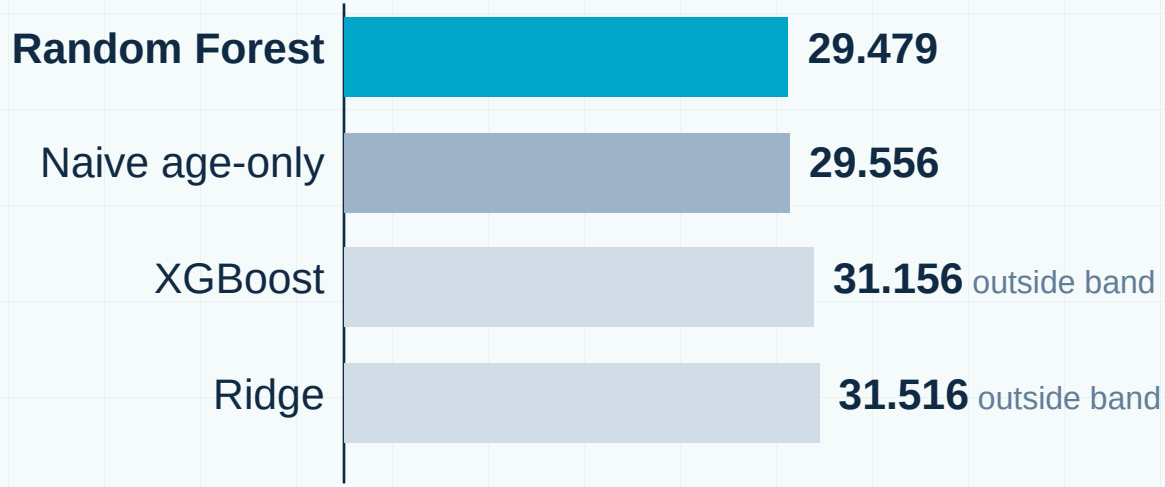
Primary target: uncapped RUL(i, t) = T_i − t

Capped at 130 cycles: sensitivity analysis only

No engine contributes rows to more than one role — so no trajectory leaks across fitting, selection, calibration and test

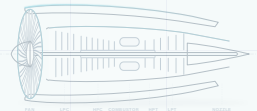
Random Forest won a near tie through the frozen rule

Mean engine RMSE · validation · cycles (lower is better)



Validation: 20 engines, 4,217 rows · rules frozen before the test · Table 5.1, §4.7 **Backup: full comparison ↗** **selection rules ↗**

One score per engine sets a deliberately wide radius

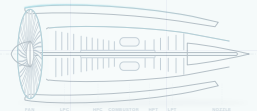


Score for each calibration engine i

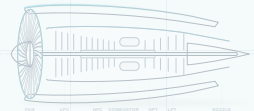
$$s_i = \max_t \left| \hat{y}(i, t) - y(i, t) \right|$$

largest absolute residual over the whole trajectory · n = 20

Nominal coverage 90% · interval $\hat{y} \pm \hat{q}$
Lower bound clipped at 0 cycles
Validity is marginal: it assumes exchangeable engines and a prespecified score



Endpoint error is moderate, but the bias is optimistic



Full coverage came with intervals too wide to schedule

Conformal score	Mean interval width · cycles · 100 endpoints	Endpoints covered	Validation trajectories fully covered (of 20)
Engine maximum FROZEN · PRIMARY	182.46	100/100	●●●●●●●●●● ●●●●●●●●●●

Wilson 95% CI 96.30–100% · median width 193.64 · 63% of lower bounds clipped at 0

Tables 5.2 and 5.7 (§5.4, §5.7) · coverage counts are observed, not guaranteed **Backup: conformal ablation ↗**

SHAP traces how the model overestimated Engine 18

TreeSHAP contributions · Engine 18 · cycles



TreeSHAP checks

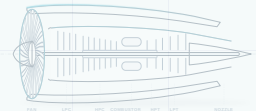
100 endpoints × 18 predictors

Additivity error < 10^{-12} cycles

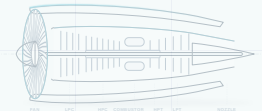
Baseline 107.84 cycles

Age ranked 1st globally (mean |SHAP| 24.76)

Local: Table 5.9 · global: Table 5.8 (§5.8; Lundberg and Lee, 2017) · ‘other 13’ = derived remainder **Backup: global SHAP ↗** **failure cases ↗**

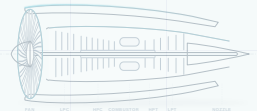


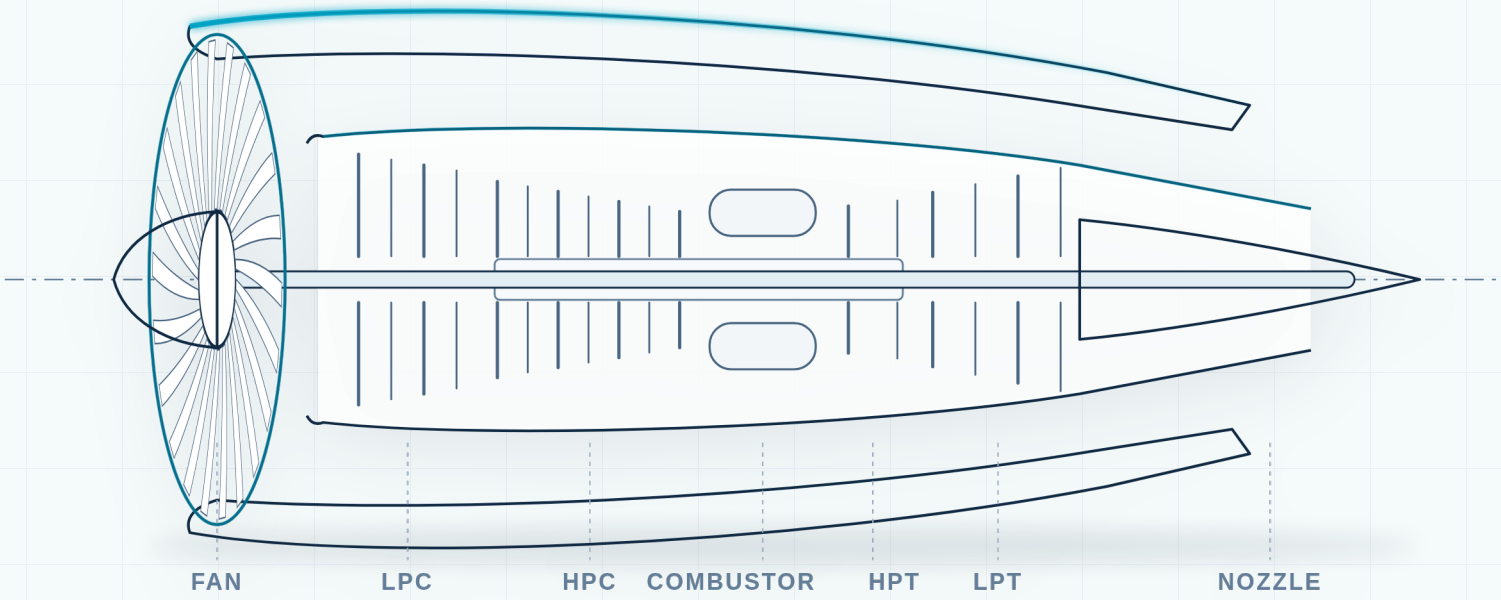
High recall came with a heavy review burden



Policy stays deterministic; the human decides

Three conclusions, bounded by clear limits





Illustrative turbofan schematic — not a validated engine simulation

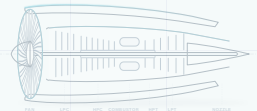
Thank you — Questions

Ranitabh Mallick · Supervisor: Dr Hiba Khalid · GISMA University of Applied Sciences

Backup slides: index ↗

BACKUP SLIDES

Evidence for questions — select a topic



B1 Research questions and answers

B2 Objectives → evidence

B3 Partitions and leakage controls

B4 Full model comparison

B5 RF and XGBoost configurations

B6 Equivalence band and PHM tie-break

B7 Conformal score and rank

B8 Bootstrap and confidence intervals

B9 Uncapped target vs capped sensitivity

B10 Feature ablation

B11 Conformal-score ablation

B12 Global SHAP and additivity

B13 Local SHAP failure cases

B14 Triage classes and denominators

B15 Endpoints vs validation rows

B16 n8n topology

B17 Decision-route evidence

B18 Audit-record fields

B19 Invalid-MODIFY limitation

B20 Reproducibility environment

B21 Provenance limitations

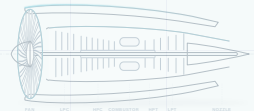
B22 Standards as design references

B23 Scope and claims boundary

B24 Verified references

B25 Repository

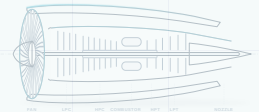
↩ Conclusions (slide 12) ↩ Questions (slide 13)



Four research questions and their bounded answers

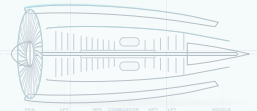
RQ	Question	Answer (§7.2)
RQ1	How accurately can lightweight ML models estimate RUL for FD001 under an engine-level, leakage-controlled design, relative to a naive age-only baseline, and which feature groups drive that accuracy?	Test MAE 18.88, RMSE 25.73, bias +10.90 cycles; PHM tie-break selected RF despite near-equal validation RMSE vs age alone
RQ2	How can engine-block conformal prediction provide RUL intervals with a transparent finite-sample rule, and what coverage–precision trade-off does it create?	q̂ 107.90 covered 100/100 endpoints, mean width 182.46; narrower scores covered only 8/20 validation trajectories
RQ3	What do SHAP explanations reveal about the selected model, and how can they be read against turbofan physics without being mistaken for causal evidence?	Exact reconstruction; age dominant; local core-speed contributions — not physical causes
RQ4	How can prediction, uncertainty and explanation be translated into CRITICAL/HIGH/MEDIUM/LOW triage with mandatory human review, and how safe and burdensome is it?	Point triage under-triaged 34/100; offline guard caught 39/39 with 81/100 reviews; prototype records three decision types

Table 1.2 (§1.3); answers §7.2 ↩ Backup index



Six objectives map to specific evidence

ID	Objective	Evidence
O1	Leakage-controlled, engine-level partitions and an auditable RUL target	Frozen 60/20/20 engine partitions, target checks, SHA-256 manifests (§4.3–4.5)
O2	Compare regression models against a clear baseline under a pre-declared policy; test feature groups	Naive, Ridge, RF, XGBoost comparison; feature ablation (§5.2, §5.6)
O3	Quantify uncertainty with a transparent, engine-aware conformal rule	Engine-block calibration, $\hat{q}$ = 107.90; conformal ablation (§4.8, §5.4, §5.7)
O4	Evaluate the frozen model once on the official test endpoints	Point, interval, lifecycle and bootstrap results, 100 engines (§5.3–5.5)
O5	Explain predictions without causal overclaiming; relate to engine physics	100 × 18 TreeSHAP matrix; local failure cases (§5.8)
O6	Specify and audit a CRITICAL/HIGH/MEDIUM/LOW triage policy for an n8n review workflow	Triage policy, classification audit, review-burden analysis (§4.10, §5.9)

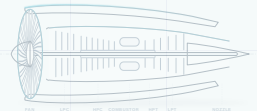


Each engine had exactly one role

Partition	Engines	Rows	Role	Prohibited use
Model training	60	12,310	Fit scaler and models; grouped CV for tuning	Any claim of held-out performance
Validation	20	4,217	Compare frozen candidates; select point model	Calibration; retuning after selection
Conformal calibration	20	4,104	One nonconformity score per engine	Model selection; method shopping
Official test	100	13,096 (100 endpoints)	Once-only final evaluation	Reselection, retraining, recalibration

Controls: fixed seed-42 engine shuffle without rerolling or stratification · scaler fitted on training engines only · Phase 6 policy froze model hash, feature order, q̂, metrics, lifecycle groups and the last-observed-row rule before the 100 official RUL values were read.

Table 4.1 (§4.5); leakage: Kaufman et al. (2012) ↩ Backup index

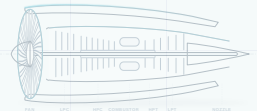


Full validation comparison: a near tie on mean engine RMSE

Model	Mean engine RMSE	Median engine RMSE	Pooled RMSE	Pooled MAE	Bias	Over-estimation	Mean PHM
Random Forest	29.479	26.458	33.073	25.324	+0.96	54.2%	83.67
Naive age-only	29.556	21.109	41.135	31.106	−19.33	33.3%	156.44
XGBoost	31.156	29.471	34.348	26.664	+1.03	53.5%	101.15
Ridge	31.516	26.076	34.295	26.251	+0.44	54.1%	157.53

Selection used mean engine RMSE with a 1% practical-equivalence band, then the mean PHM penalty. Other columns are descriptive. The age rule's much lower median engine RMSE and strongly negative bias show it errs pessimistically.

Table 5.1 (§5.2) · 20 validation engines, 4,217 rows ↩ Backup index

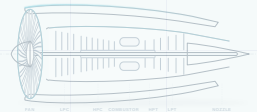


Frozen configurations from grouped cross-validation

Model	Role	Training-only selection	Frozen configuration
Naive age-only	Non-sensor baseline	Median training lifetime (195 cycles)	$\hat{y} = \max(195 - \text{time_in_cycles}, 0)$
Ridge	Regularised linear comparator	11 α values, grouped 5-fold CV	$\alpha = 0.0001$ (grid boundary)
Random Forest	Non-linear bagged trees	18 candidates, grouped 5-fold CV, 150 trees per fold	depth 8, min. leaf 10, max_features 0.5, 500 trees, seed 42
XGBoost	Regularised boosted trees	18 candidates, grouped 5-fold CV	depth 3, learning rate 0.03, min. child weight 5, 400 rounds, subsample 0.8, colsample 0.8, $\lambda = 5$

Ridge standardisation fitted on training engines only; tree models used raw inputs; outputs clipped at 0; no retuning after configurations were fixed.

Selection: equivalence band first, asymmetric penalty second



Rule 1 — practical equivalence

Candidates whose mean engine RMSE lies within 1% of the best are treated as equivalent.

RF 29.479 vs naive 29.556: ratio 1.0026 → within 1%

XGBoost 31.156 and Ridge 31.516: outside the band

Rule 2 — tie-break

Lowest mean PHM penalty among equivalent candidates: RF 83.67 vs naive 156.44.

PHM 2008 penalty, e = predicted − actual

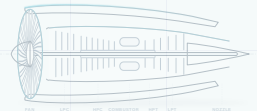
$e \geq 0$ (late): $\exp(e / 10) - 1$

$e < 0$ (early): $\exp(-e / 13) - 1$

Late, optimistic errors are penalised more steeply than early ones.

Ratio 29.556 / 29.479 is derived from Table 5.1. Both rules were fixed before validation results were read.

Conformal score: definition, rank and assumptions



Score

$s_i = \max_t |\hat{y}(i, t) - y(i, t)|$ for each of n = 20 calibration engines

Rank

Finite-sample rank $\lceil (n + 1)(1 - \alpha) \rceil = \lceil 21 \times 0.90 \rceil = 19$ → 19th of 20

$\hat{q}$ = 107.8965 cycles (reported 107.90)

Interval

$[\max(\hat{y} - \hat{q}, 0), \hat{y} + \hat{q}]$; clipping changes reported width

Assumptions and caveats

Coverage is marginal under exchangeable calibration and future engines

Not guaranteed per engine, under distribution shift, or as a safety certificate

Calibration covered 19/20 engines by construction — not independent validation

20 engines → coarse order statistic; $\hat{q}$ sensitive to one engine

Official endpoint results with bootstrap intervals

Measure	Value	Interpretation
MAE	18.88 (95% CI 15.56–22.48)	Average absolute endpoint error
RMSE	25.73 (CI 21.33–30.15)	Larger than MAE: a minority of large errors
R^2	0.617 (CI 0.42–0.75)	Meaningful but incomplete variance explained
Bias	+10.90 (CI 6.47–15.57)	Systematic overestimation; interval excludes zero
Median absolute error	14.19	Half of engines within about 14 cycles
Over- / underestimation	69% / 31%	Overestimation dominates
Within 10 / 20 / 30 cycles	43% / 62% / 75%	Descriptive accuracy bands
PHM penalty, total / mean	10,108.02 / 101.08	Inflated by late predictions

The bootstrap quantifies sampling uncertainty of these same 100 predictions — not performance on new engines or fleets.

Table 5.2 (§5.3) · 100 engines · 5,000 engine resamples of the single evaluation ↩ Backup index

Capping changes the question, not just the score

Lifecycle stage	MAE uncapped	MAE capped	Bias uncapped	Bias capped
Critical, 0–30	7.48	7.32	+4.87	+4.68
Near-term, 31–60	19.96	18.91	+13.09	+11.95
Mid-life, 61–130	29.02	17.73	+13.72	−0.81
Early life, > 130	31.03	58.01	−16.29	−58.01

Against actual cycles

Capped-130 validation RMSE 44.04 vs 33.07 uncapped

Capped bias −19.85; mean PHM penalty 1,440.27

Uncapped RUL is primary; capped is sensitivity only.

Table 5.5 (§5.5) · validation rows, actual-RUL scale ↩ Backup index

Age dominates; dropping s9/s14 is only a hypothesis

Variant	N features	Val. MAE	Pooled RMSE	Bias	Mean PHM	Engine MAE
A0 Full model	18	25.32	33.07	+0.96	83.7	24.93
A1 Without time_in_cycles	17	31.03	41.16	+1.77	1,436.3	30.18
A2 Sensors only	15	31.02	41.14	+1.72	1,389.4	30.17
A3 Age only (Random Forest)	1	28.79	36.72	−1.80	93.0	28.28
A4 Without operational settings	16	25.40	33.14	+0.99	84.7	25.00
A5 Without sensor_9 and sensor_14	16	24.03	31.76	−1.44	60.1	23.79
B0 Naive age-only rule	1	31.11	41.14	−19.33	156.4	28.73

A5 lowered pooled RMSE about 4.0% in a single exploratory comparison on reused validation engines — it does not change the frozen model or establish general harm.

Table 5.6 (§5.6) · post-closure · 60 modelling, 20 reused validation engines, no retuning ↩ Backup index

Narrower conformal scores lose whole-trajectory coverage

Score	q̂	Val. row coverage	Val. engines fully covered	Test endpoint coverage	Test mean width	Test misses below lower bound
C0 Engine maximum (frozen)	107.90	100%	20/20	100%	182.46	0
C1 Row-level	57.70	89.7%	8/20	96%	106.34	4
C2 Engine mean	54.32	88.2%	8/20	96%	100.72	4

Selecting C1 or C2 on these already examined engines would import post-closure test knowledge. Replacement needs fresh engine groups and a prespecified comparison.

Global SHAP ranks age first, then HPC pressure

Rank	Feature	Physical meaning	Mean |SHAP|	Mean signed
1	time_in_cycles	Engine age	24.76	−16.57
2	sensor_11	Ps30, HPC outlet static pressure	12.87	−1.78
3	sensor_4	T50, LPT outlet temperature	7.45	−0.68
4	sensor_9	Nc, core speed	6.05	+0.40
5	sensor_14	NRc, corrected core speed	5.30	+0.55
6	sensor_12	phi, fuel flow / Ps30	4.39	−0.45
7	sensor_7	P30, HPC outlet pressure	3.59	−0.77
8	sensor_15	BPR, bypass ratio	2.05	−0.61
9	sensor_13	NRf, corrected fan speed	1.03	−0.45
10	sensor_8	Nf, fan speed	0.85	−0.48

Checks

Additivity / fidelity: max discrepancy < 10^{-12} cycles

Units: cycles of predicted RUL

Correlated channels can exchange credit; sensor names follow the literature mapping

Non-causal

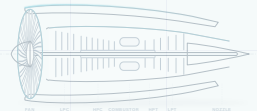
Table 5.8 (§5.8) · 100 official endpoints × 18 predictors · baseline 107.84 cycles ↩ Backup index

Four local cases: two under-triaged, two large errors

Engine	Age	True RUL	Predicted	Largest contributions (cycles)
41	123	18	60.0	age −29.2; T50 (s4) −11.5; phi (s12) −7.9; Nc (s9) +4.3; NRf (s13) −3.0
18	133	28	64.5	age −28.1; Ps30 (s11) −12.2; T50 (s4) −11.4; Nc (s9) +5.6; NRc (s14) +4.6
15	76	83	170.5	Ps30 (s11) +18.3; NRc (s14) +9.5; T50 (s4) +8.2; Nc (s9) +7.8; phi (s12) +7.5
1	31	112	184.2	age +34.7; Ps30 (s11) +14.3; T50 (s4) +7.0; NRc (s14) +6.0; Nc (s9) +5.2

Engines 41 and 18 (truly CRITICAL, point class MEDIUM): core-speed inputs partly counteracted degradation-related negative contributions. Engine 15's measured sensors looked healthy with 83 cycles left. These describe model behaviour, not a diagnosed physical cause.

Table 5.9 (§5.8.2) · baseline 107.84 cycles · contributions in cycles ↩ Backup index



Triage classes are stipulated; Rule 2 is offline only

Class	Rule 1 on predicted ŷ	Proposed action (for reviewer approval)	Review requirement
CRITICAL	ŷ ≤ 30 cycles	Immediate inspection planning; consider removal from service	Mandatory human review
HIGH	30 < ŷ ≤ 60	Schedule inspection and parts within the planning horizon	Mandatory human review
MEDIUM	60 < ŷ ≤ 130	Increased monitoring; plan at the next maintenance opportunity	Routine queue unless Rule 2
LOW	ŷ > 130	Continue routine monitoring	Routine queue unless Rule 2

Rule 2 (post-closure, offline): refer when the conformal lower bound is ≤ 30 cycles. Not implemented as n8n routing. Denominators: 100 endpoints — 25 truly CRITICAL, 39 truly CRITICAL-or-HIGH; point-only CRITICAL recall 64%.

Endpoints and validation rows answer different questions

Measure	Endpoints (n = 100 engines)	Validation rows (n = 4,217, 20 engines)
Point-only: CRITICAL recall	64%	80.5%
Point-only: CRITICAL or HIGH recall	82%	77.6%
Point-only: under-triage rate	34%	19.8%
Two rules: truly CRITICAL routed to review	100%	100%
Two rules: truly CRITICAL or HIGH routed	100% (39/39)	99.8% (1,218/1,220)
Two rules: share of all cases reviewed	81%	61.1%
Reviewed cases truly CRITICAL or HIGH	48.1%	47.2%
Routed by Rule 2 only	47%	35.6%

Validation rows are repeated cycle observations of the same 20 engines — not independent. Both columns are retrospective under simulated labels; neither is a field safety rate.

Exported 22-node n8n workflow (THESIS_HITL_AGENTIC_V2)



Groups

Fixed PROTOTYPE_TEST input

Evidence and provenance checks

DEMO_POLICY_V1 (STIPULATED_RESEARCH_ASSUMPTION)

Bounded local narration (Ollama, Qwen3 1.7B, temperature 0)

Human form → APPROVE / REJECT / MODIFY

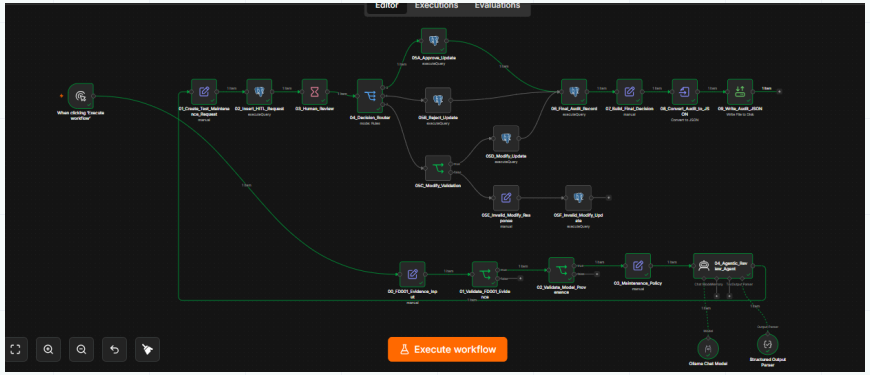
PostgreSQL status + JSON audit export

Connections show available routes; exported audit records show saved decisions. Neither establishes a live notebook feed, offline interval-guard routing or field reliability.

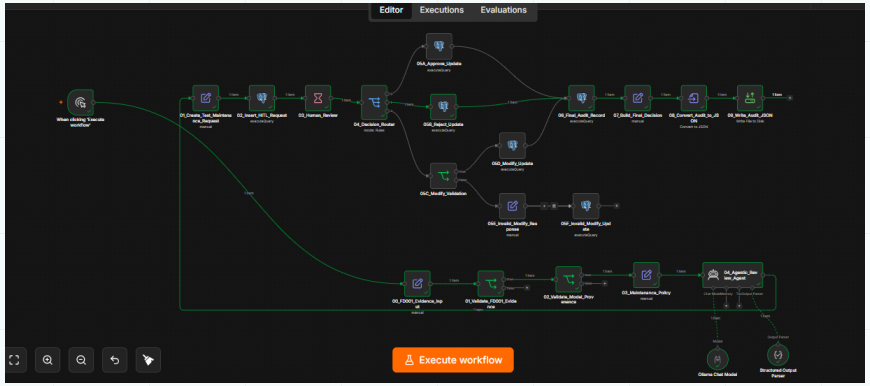
Figure 4.7 (§4.10.4) · author's n8n editor screenshot — topology, not a deployment ↩ Backup index

Three human decision types were executed and exported

Branch	Observed result	Evidence
APPROVE	Executed twice; APPROVED, override NO	Two audit records
REJECT	Executed; REJECTED, override YES, final recommendation empty	Audit record
MODIFY (valid)	Executed; MODIFIED, override YES, revised recommendation stored	Audit record
MODIFY (invalid)	Editor trace reaches invalid response and update nodes	Trace only; no audit JSON



APPROVE route highlighted (editor execution)



REJECT route (editor execution)

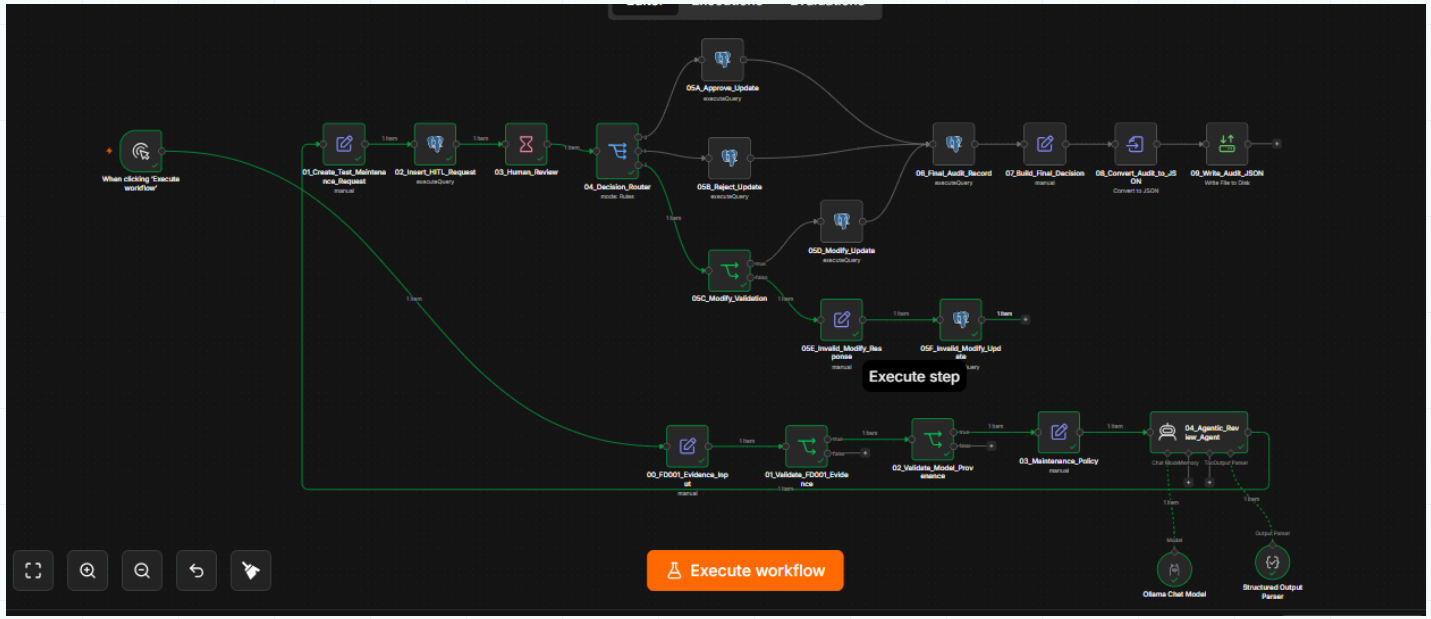
Audit records link policy, explanation and decision

Field group	Recorded content (MODIFY example)
Request	Request id; asset id; predicted RUL 18 cycles; priority CRITICAL
Policy	DEMO_POLICY_V1; STIPULATED_RESEARCH_ASSUMPTION; stage critical_0_to_30; flag LOWER_BOUND_CLIPPED; rule reason
Explanation	SHAP drivers + constrained agent summary, evidence list, limitations, ‘human review required’
Human decision	MODIFY; comment; revised recommendation; status MODIFIED; override YES
Timestamps	Creation, review and completion

Record	Decision	Override
HITL-1790096772107	APPROVE	NO
HITL-1790108329010	APPROVE	NO
HITL-1790097014976	REJECT	YES
HITL-1790094775152	MODIFY	YES

A fifth, earlier placeholder APPROVE run (HITL-20260909-214406-369) is kept separate from these four agent-assisted records.

Invalid MODIFY: a route trace, not a persisted-state proof



Invalid-MODIFY response and update nodes highlighted (editor execution)

What is and is not shown

Expected: fail closed when no revised recommendation is supplied

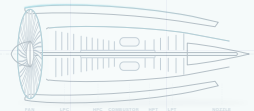
Shown: the editor trace reaches the invalid response and update nodes

Not shown: a returned database row or exported audit JSON of the final state

Executed environment and independent reproduction

Component	Version / platform
Python (original)	3.13.15 (Google Colab, Linux)
Python (Phase 8 reproduction)	3.12
NumPy / pandas	2.1.3 / 2.2.3
scikit-learn	1.6.1
XGBoost	3.4.1
SHAP	0.52.0
Matplotlib / seaborn	3.10.0 / 0.13.2
Workflow layer	n8n + PostgreSQL + local Ollama chat model; fixed evidence payload

The Phase 8 reproduction restored the 60/20/20 partition with the same library versions and matched the recorded validation, calibration, test and SHAP values exactly. The two Python environments are reported separately.



Numbers reproduce; earlier history cannot

Reproduced

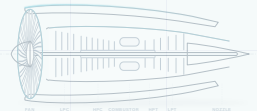
Independent Phase 8 run restored the partition and matched $\hat{q}$ and the principal validation, calibration and test metrics. The selected model hash stayed stable.

Not reconstructable

Five legacy Phase 5–6 artefact hashes differ from current archive bytes. The number of historical test executions is not fully verifiable, so post-closure analyses are treated as descriptive characterisation.

Required response

Version control with signed, immutable releases from ingestion onward; fresh data for confirmation.

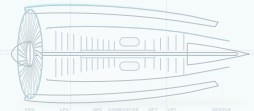


Standards informed the design — no compliance is claimed

Standard or framework	Application in this dissertation
EN 13306:2017	Definitions of fault, failure, condition-based and predictive maintenance
ISO 17359:2018	Monitored parameters and alert criteria that trigger review
ISO 13374-1:2003	Pipeline blocks: acquisition → state detection → health → prognosis → advisory
ISO 13381-1:2015	RUL definition; confidence stated alongside a prognosis
IEEE Std 1856-2017	Sensing, health assessment, prognostics and decision support as distinct capabilities
ISO 55000:2014; ISO 31000:2018	Risk-based prioritisation behind the four classes
NIST AI RMF 1.0; ISO/IEC 42001:2023	Governance via frozen policies, documented limitations, version fingerprints
EU AI Act, Art. 14 (2024)	Reviewer authority to approve, reject, modify or defer; no autonomous approval

What the framework can and cannot do

The framework CAN	The framework CANNOT
Estimate RUL for FD001-like engines from 18 cycle-level inputs	Transfer to other conditions, fault modes (FD002–FD004), engine types or real fleets without new validation
Attach a 90% interval calibrated on whole engine trajectories	Guarantee coverage for an individual engine, under distribution shift, or as a safety certificate
Show which inputs moved a prediction, and by how much	Identify the physical cause of degradation or the failing component
Assign stipulated classes; evaluate an offline uncertainty-based review rule	Define an approved maintenance window, cost-optimal action or airworthiness decision
Record proposals, reviewer decisions and versions for audit	Provide tamper-proof logging, access control, uptime or cybersecurity guarantees
Support a qualified human reviewer	Replace the reviewer or approve maintenance autonomously



Key references (Harvard)

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Dissertation repository (private)

github.com/mranitabh/GISMA-hitl-dissertation-predictive-maintenance

Private dissertation repository — examiners may need to request access.

Contents relevant to the viva

executed_notebook/ — executed implementation notebook (HTML export)

workflow/ — n8n workflow exports

audit_examples/ — sanitised HITL audit JSON records

screenshots/ — workflow canvas, form and route traces

schemas/, config/ — request/response schemas and threshold policy

presentation/ — this deck, its source, script and verification report